<u>Automatic Tree Tagger User Manual</u>

Author: Daniel Butt, NTP 2024
dbutt7@uwo.ca

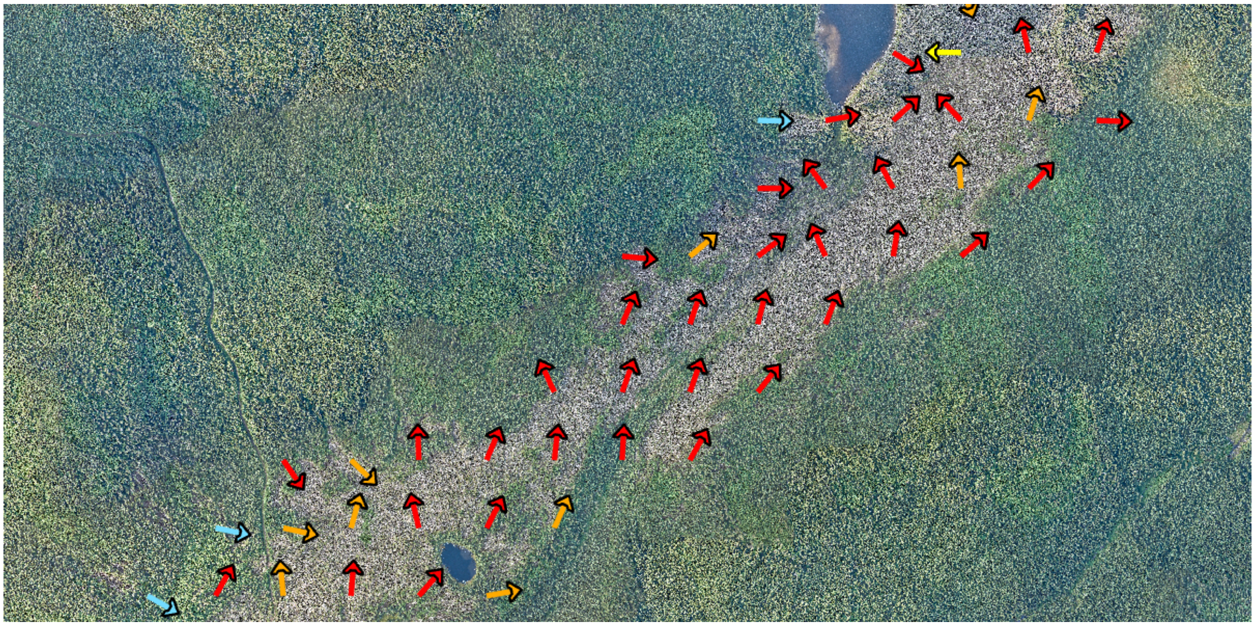


Table Of Contents

1. Overview

This software allows for the automatic detection, "tagging" (fitting a line), and direction analysis of fallen trees in large scale aerial imagery. It was designed for the Northern Tornadoes Project, mainly for analysis of fallen trees due to severe storms such as a Tornadoes. The software leverages machine learning algorithms primarily written in C++ code, however it is intended that a user will utilize the software through an ArcGIS Pro add-in built in C# with the ArcGIS SDK. Currently, the software only supports analysis of image files (Image file types supported by OpenCV) which have been georeferenced to have both coordinates and an image pixel to coordinate scale. For more information on the algorithms used in this software please see the associated paper.

1.1. Recommended Minimum System Specs

CPU: i5/Ryzen 5 - i7/Ryzen 7 made after 2017, but any half decent modern cpu should work

Ram: 16GB. Note: main test systems had at least 32GB

GPU: NVIDIA, gtx 1060 6GB or better. Note: gpu's tested, 1070, 3070/80, 4080/90, rx 6800

Storage: nvme ssd or at least ssd recommended, but a standard hard drive will work

1.2. Installation

Requires ArcGIS Pro 3.3, (should be compatible with later versions). Under the releases tab, download the latest TreeTaggerModule.esriAddinX file. Open the file and click install.

2. User Instructions

2.1. Important Notes

Primarily the machine learning models were trained and tested on 5cm imagery and it is not recommended to run the model on lower quality imagery (10cm, 20cm, etc). If you attempt to run the software on non-5cm imagery, a copy of the images will automatically be re-scaled to match 5cm imagery as the models expect a tree to look a certain way and be a certain size.

Using non-5cm imagery may also result in runtime errors. This has not been thoroughly tested. Imagery should be in a meters based not degrees based spatial reference format (Mercator EPSG:3857 vs WGS84 EPSG 4326). The spatial reference of an image can be changed in ArcGIS Pro using the "Project Raster" analysis tool.

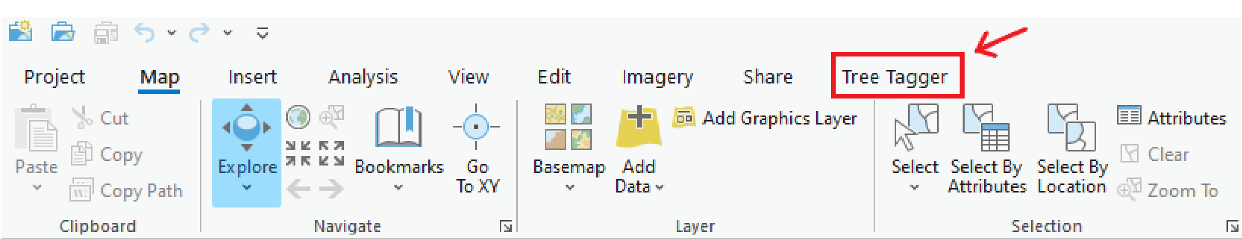
Recommend image file formats: Uncompressed TIF, PNG or JPG. Compressed TIF formats may or may not work due to issues with the OpenCV library used for image reading.

Using very large individual images (> ~2GB in uncompressed size) will probably cause your PC to crash due to RAM constraints. You can split large images into smaller sections using the "Split Raster" ArcGIS Pro analysis tool.

In order for the software to run optimally (speed of analysis), it is strongly recommended to run the software using a GPU to speed up the machine learning model's prediction time. This software utilizes DirectML, which should automatically use any recent GPU. Also, having your imagery on a fast drive (nvme ssd > ssd > hard drive) will greatly improve image loading times.

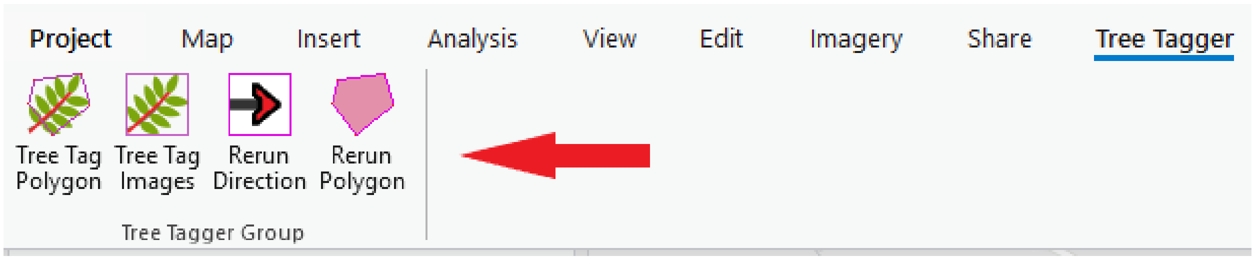
2.2. Basic Usage

Once installed correctly, you should now see an additional tab in the top ribbon of any ArcGIS Pro project labelled "Tree Tagger".



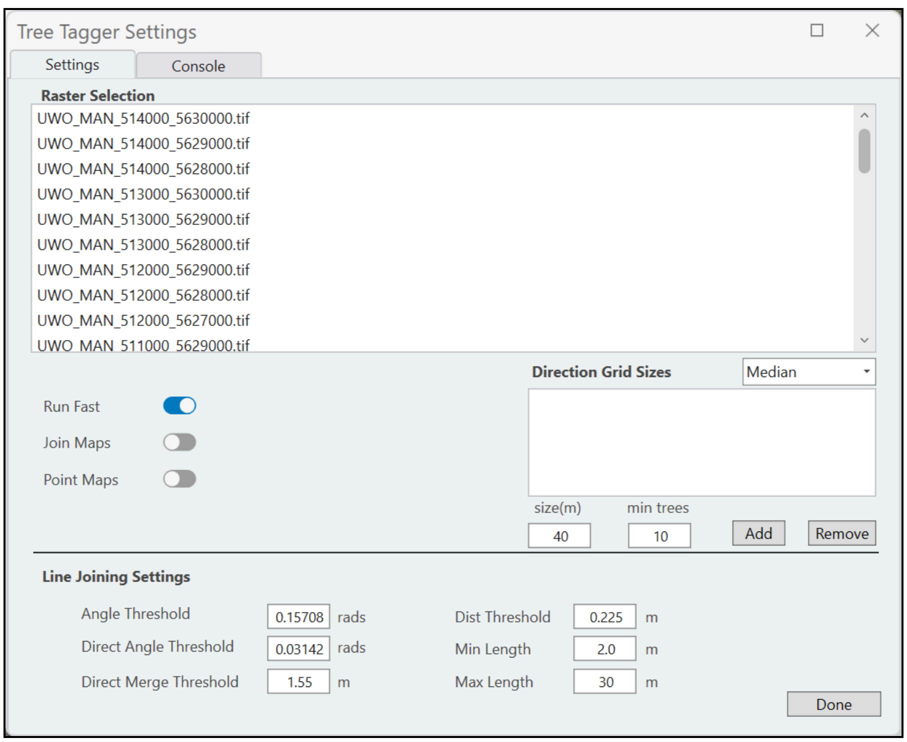
Selecting this tab will display two main options for running the program, either "Tree Tag

Polygon" or "Tree Tag Image". As well as, two options for rerunning directions or high density polygons, which are discussed later.



If you know the specific image(s) you wish to process, or you plan to process the entire event, it it recommended that you use Tree Tag Images. If you only want to process a specific region, for example the damage track, you can instead select a polygon region using the Tree Tag Polygon option.

Once you have finished selecting a polygon region, or if you select Tree Tag Images, the Tree Tagger settings menu will open.

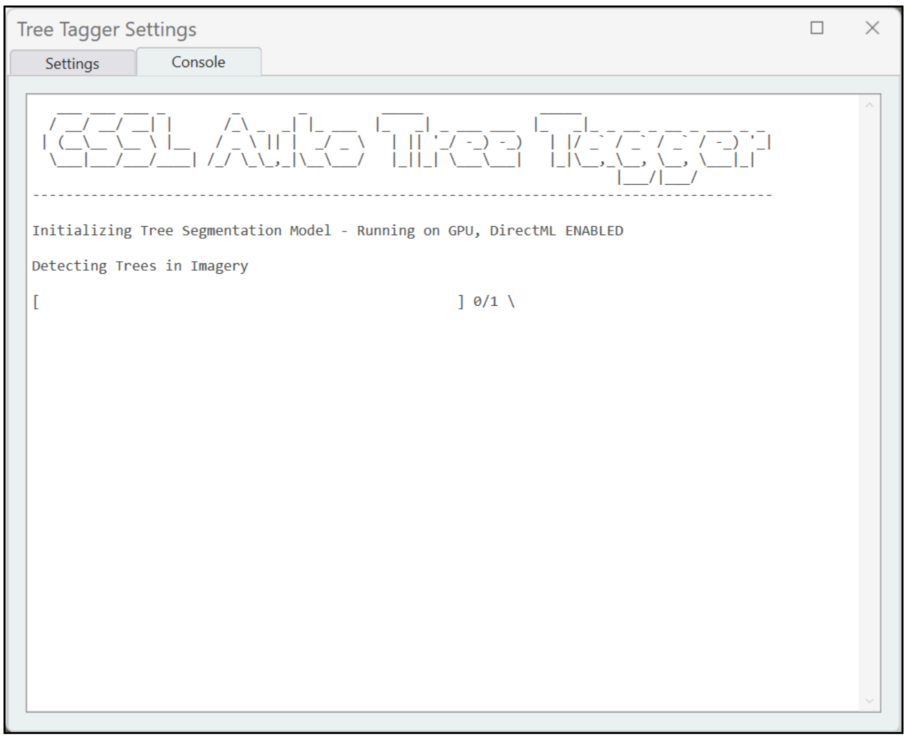


From here you can select the specific images you wish to process using the "Raster Selection" menu and change any other settings required (see the various settings sections). If you selected a polygon region, the images will be selected for you automatically and the Raster Selection menu will be disabled. You must input at least one direction grid size by using the "size" and "min trees" boxes and then pressing "Add".

Tips:

- Clicking on an image in ArcGIS will tell you the image's name
- You can use shortcuts for selecting multiple images in the Raster Selection menu
    * Hold ctrl to select multiple images
    * Select an image, then hold shift and select another image to select a range
    * Press ctrl + A to select all images

Once you have finished selecting images and changing settings, press "Done" to start the analysis. In the "Console Tab" you should see something like the following indicating the progress of the analysis.



While the program is running, you most likely won't be able to use certain features of ArcGIS and it is not recommended to perform any other computationally intensive tasks in the background. The analysis this program performs is very computationally intensive and may take a long time to run depending on your computer specifications, how many images were selected, the size of the images, and the number of trees detected.

Once the analysis has completed ArcGIS should automatically load the results. You should now see that a group called Tree Tagger was created containing 5 new raster layers labelled "Lines", "Vectors", "Directions", "Refined_Lines", and "Outlines". The results are saved as shape files (and csv in the case of directions) and can be located in a created folder labelled "Tree Tagger" in your ArcGIS project's directory.

In the event the program crashes, you should receive an error message describing the error in the console window. Following a crash, if some of the analysis has finished, you can find the results in the located in the same "Tree Tagger" folder in your ArcGIS project's directory.

2.2.1. Lines

The Lines shape file contains the coordinates of the end points of all lines detected from the original analysis of the imagery.

2.2.2. Vectors

The Vectors shape file contains the coordinates of the end points of all predicted line directions (excluding the lines predicted to be false positives or inconclusive). It can be used to see the predicted directions for individual trees, as well as, used to rerun directions at either differing grid sizes, or after manual removal of false positives / unimportant sections.

2.2.3. Refined_Lines

The Refined_Lines shape file contains the end points of all refined lines. The lines were refined using the analyzed line directions and removal of inconclusive lines (lines which are either a detected false positive or whose direction couldn't be extracted).

2.2.4. Directions

The Directions shape file contains the area averaged treefall directions as either lines or points depending on your selection. For lines, make sure you select arrow symbols which are drawn from first to last end point in ArcGIS or all the directions will be backwards. For points, you will need to set the rotation based on the shapefile's "Angle" field. Once you have finished applying your desired symbology (formatting) of the direction vectors, you can save a copy of the directions shape file as a layer and use the "Apply Symbology From Layer" ArcGIS tool to copy the symbology from the layer file to a future shape file. Also if you select "Join Maps" it will join all the direction grid sizes into a single shapefile. From here it is recommended to filter the vector sizes using scale based display filters so only certain vectors appear at certain scales using the shapefile's "Layer" field.

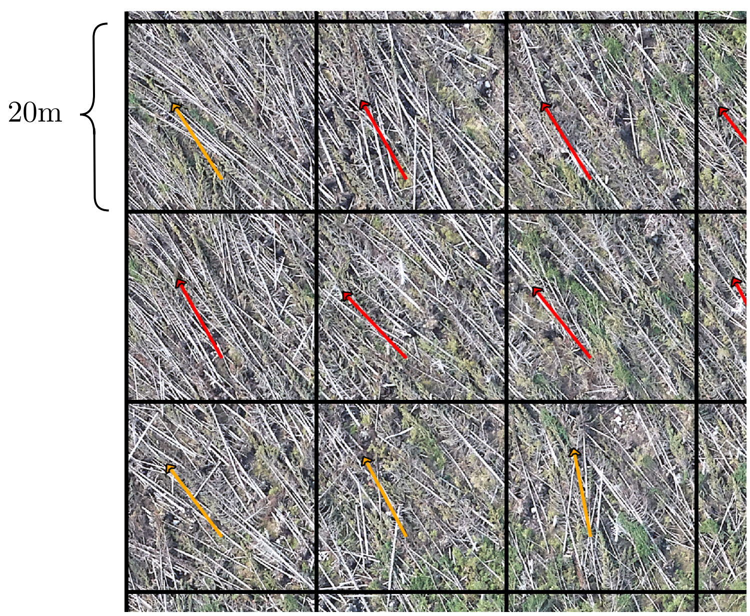
2.2.5. Outlines

The Outlines shape file contains the points of polygons which enclose the direction lines or points in order to outline the areas of damage

2.3. Direction Grid Settings

The direction grid settings refer to the sizes of the grid squares in which the individual tree directions will be averaged, as well as, the minimum number of found trees in a grid square or that square to be considered. For each size, the analyzed images are split into a grid of the specified size. Next, all the tagged trees which have predicted directions, will be sorted into the grid squares based on the location of their line's mid-point. From here an area average of all the tree directions is calculated based on the selected method (median, dual median, dual medoid) for each grid square that exceeds the minimum number of trees required. See paper for more details.

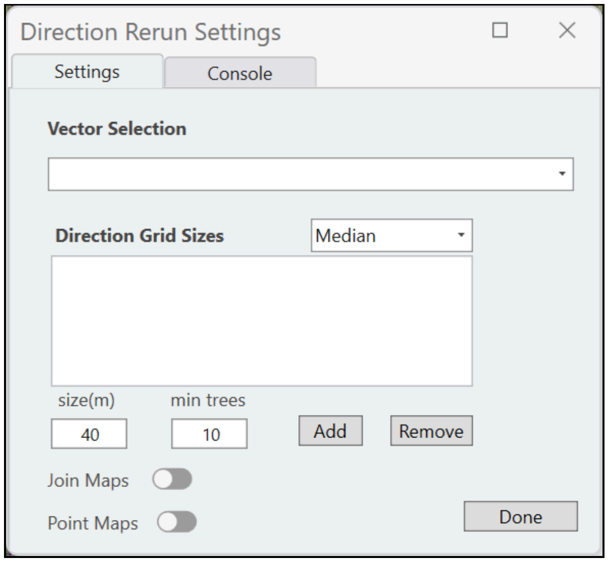
Generally it is not recommend to set a size smaller than 10m as there may not be enough trees to average per grid square, leading to inaccurate results. Moreover, it is not recommended to exceed a size larger then that of your imagery (say your images were 1km x 1km and you picked a size of 2km x 2km) as the direction vectors may not scale appropriately. If your event is very large or very small, you should adjust the sizes accordingly.



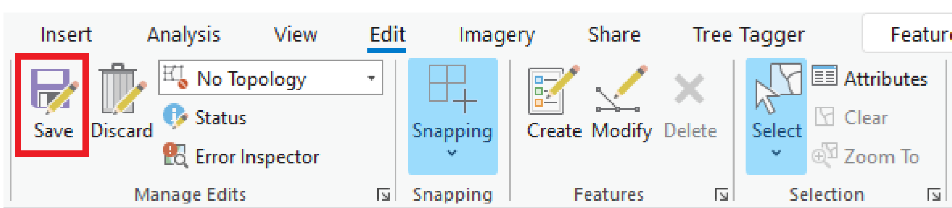
3. Re-Running Directions / Outlines

The rerun directions / outline options in the Tree Tagger ribbon allow you to rerun the direction averaging process or outlining areas should you wish to uses differing settings or manually remove sections from the original output.

In order to rerun, you must select the generated vector shape file and enter the desired direction grid sizes. Make sure you click add to add the new direction size.



Any manual removal of false positives or undesired areas must be done to the vector shape file. Make sure only the vector shape file is selected when removing elements to ensure you don't accidentally remove parts of another shape file. Also make sure you save the shape file after editing.



4. Troubleshooting

Have you tried turning it off and then on again? :)

In general if the Tree Tagger software encounters an error, you should be given an error message describing the error and potential problems. If the error message doesn't make any sense, or isn't really human readable, then most likely the error is with ArcGIS and can be solved by restarting ArcGIS.

4.1. Done button not working

Make sure you have selected an image to process or that your selected polygon region intersects an image that can be processed (see Overview for supported file types).

4.2. Direction vectors are all pointing the opposite direction

Most likely you have selected an ArcGIS shape file symbol which draws vectors (or lines) from last to first endpoint, instead of the required first to last endpoint. If this is not the case then a catastrophic error as occurred, please check your images files are of supported types and their coordinates and coordinate system make sense. Note, this software has not been rigorously tested with every supported coordinate system ArcGIS offers.

4.3. Crashed or won't run with no clear error message

If it won't start at all, try saving and restarting ArcGIS. If this is the first time you're using the Tree Tagger software on your specific computer, you may want to double check you have installed the software correctly (see installation guide). Otherwise, see the "Done button not working" section above. If this doesn't fix the problem you may have restart your computer. If this still doesn't fix the problem, try reinstalling Tree Tagger.

If it crashed during the analysis process and you didn't receive a helpful error message, the following is a list of potential issues,

- ArcGIS itself has had an error and maybe even crashed, try restarting ArcGIS

- Your ArcGIS version is less than 3.3 or you're missing the .Net Desktop Runtime 8.0 that is required for ArcGIS 3.3

- Your images are not of supported file types, of an irregular scale/coordinate system, not accessible by Tree Tagger (in use by another program), or corrupted

- Your trying to process a very large event and your computer doesn't have enough ram, try running on fewer images / smaller images

- Your storage drive is full and Tree Tagger can't save the result files.

If non of these seem to be the issue or you're really stuck, feel free to send me an email at dbutt7@uwo.ca.